



Reduced Wide Local Excision Margins are Associated with Increased Risk of Relapse and Death from Merkel Cell Carcinoma

Neal Andruska, MD, PhD¹, Lily Mahapatra, MD, PhD², Randall J. Brennen, MD, PhD¹, Jason T. Rich, MD³, Brian C. Baumann, MD¹, Leigh Compton, MD, PhD^{2,4}, Wade L. Thorstad, MD¹, and Mackenzie D. Daly, MD¹

¹Department of Radiation Oncology, Siteman Cancer Center, Washington University School of Medicine, Saint Louis, MO; ²Department of Pathology and Immunology, Siteman Cancer Center, Washington University School of Medicine, Saint Louis, MO; ³Department of Otolaryngology, Siteman Cancer Center, Washington University School of Medicine, Saint Louis, MO; ⁴Division of Dermatology, Siteman Cancer Center, Washington University School of Medicine, Saint Louis, MO

ABSTRACT

Introduction. Current recommendations regarding the size of wide local excision (WLE) margins for Merkel cell carcinoma (MCC) are not well established.

Methods. WLE and pathologic margins were respectively reviewed from 79 patients with stage I or II MCC, who underwent WLE at Washington University in St Louis from 2005 to 2019. Outcomes included local recurrence-free survival (LRFS), regional recurrence-free survival (RRFS), distant recurrence-free survival (DRFS), disease-free survival (DFS), and disease-specific survival (DSS).

Results. Thirty-two percent of patients received adjuvant radiotherapy (aRT). At 1 year, DFS was 51.3%, 71.4%, and 87.8% for patients with WLE margins < 1 cm, 1–1.9 cm, and ≥ 2 cm, respectively ($p = 0.02$). At 3 years, the DSS was 57.7%, 82.6%, and 100% for patients with WLE margins < 1 cm, 1–1.9 cm, and ≥ 2 cm, respectively ($p = 0.02$). Multivariable Cox analysis demonstrated that every 1-cm increase in WLE margins was associated

with improved RRFS [hazard ratio (HR) = 0.28, 95% confidence interval (CI): 0.11–0.75], DRFS (HR 0.30, CI 0.08–0.99), DFS (HR 0.42, CI 0.21–0.86), and DSS (HR 0.16, CI 0.04–0.61). WLE and pathologic margin size were moderately-to-strongly correlated ($r = 0.66$). Close or positive pathologic margins (< 3 mm) were associated with reduced DRFS (HR 6.83, CI 1.80–25.9), DFS (HR 2.98, CI 1.31–6.75), and DSS (HR 3.52, CI 1.14–10.9).

Conclusion. Reduced WLE and pathologic margins were associated with higher risk of relapse and death from MCC. Larger WLE margins are important in populations with lower rates of adjuvant radiation.

Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine skin cancer with an estimated 2500 cases per year in the United States.¹ The incidence has tripled over the past 20 years, and the impact of MCC on public health is growing.¹ Prognosis is closely linked to the stage at presentation with stage I and II disease representing localized disease, stage III representing nodal disease, and stage IV representing distant metastatic disease. While a combination of surgery and radiation can be curative for stage I–III disease, relapses are relatively common.

In patients with localized disease, removal of the primary tumor with wide local excision (WLE) is recommended. The National Comprehensive Cancer Network (NCCN) guidelines currently recommend 1- to 2-cm resection margins to the investing fascia of muscle when clinically feasible.^{2–5} Yet, the literature remains sparse in

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M. D. Daly, MD

e-mail: mdaly@wustl.edu

terms of studies supporting a specific margin recommendation, and surgeons will typically use their discretion when selecting a margin width, with the goal of minimizing postoperative morbidity without compromising oncologic outcomes. Two prior retrospective studies reported no significant difference in local recurrence rates between patients who had final surgical margins of 1 and 2 cm.^{6,7} Given that 2-cm resection margins often require a skin graft or flap reconstruction for definitive closure, there is a trend toward reducing WLE margins.

MCC is associated with high risk of local recurrence following surgical excision, with up to 40% of all patients developing local recurrence even with wide margins.⁸ Adjuvant radiation therapy (aRT) to the site of the primary tumor is often recommended. A higher risk of tumor recurrence is associated with chronic immune suppression, larger tumors (diameter > 1 cm), the presence of lymphovascular invasion (LVSI), and pathologically positive surgical margins. When one of these risk factors is present, NCCN guidelines recommend adjuvant radiation.²⁻⁵ Approximately 52% of patients with localized MCC nationwide receive aRT, but there is wide variability in institutional practices.⁹ There is limited data on whether larger WLE margins are important in patients that do not receive aRT.¹⁰

MATERIALS AND METHODS

This study was approved by the Institutional Review Board at Washington University School of Medicine. Patients with a diagnosis of MCC between 2005 and 2019 who underwent definitive resection of their primary tumors were retrospectively identified. All patients underwent imaging to rule out locally advanced or metastatic disease (70% PET/CT and 30% CT alone). WLE margin size was determined from the operative report, and patients were excluded from further analysis if their operative report failed to clearly document resection margins. Pathologic margin size was determined following surgery by a board-certified dermatopathologist.

Endpoints

Primary endpoints included local recurrence-free survival (LRFS), regional recurrence-free survival (RRFS), distant recurrence-free survival (DRFS), disease-free survival (DFS), and disease-specific survival (DSS). Local recurrence was defined as a tumor recurring within 2 cm of the excision scar or graft site. Regional recurrence was defined as a tumor recurring in the primary regional lymph nodes, or anywhere between the primary tumor and draining regional lymph node basin that was more than

2 cm from the excision scar or graft. Distant recurrence was defined as a tumor recurring in distant lymph nodes, visceral organs, soft tissue, or bones. DFS was defined as freedom from any cause of recurrence. DSS was defined as death from MCC.

Statistical Analysis

Chi square tests were performed for categorical variables, analysis of variance (ANOVA) was performed for continuous variables, and Kruskal–Wallis rank sum tests were performed for a categorical and nonparametric continuous variable. Pearson correlation was used to test for strength of association between two continuous variables. Survival analysis was performed using Kaplan–Meier plots and compared using log-rank tests. Univariable and multivariable hazard ratios were estimated using Cox regression analysis. Covariables with a *p* value less than 0.10 in univariable analysis were further evaluated in multivariable analysis. Overall model significance was assessed using the Wald statistic. A patient was excluded from multivariable analysis if data for one or more variables was missing. All analysis was performed in R software (Vienna, Austria), version 3.6.2. *p* values < 0.05 were considered statistically significant.

RESULTS

Patient Demographics and Clinicopathologic Characteristics

Seventy-nine patients with clinically localized disease were evaluated. Patients underwent CT (30%) or PET/CT (70%) imaging, which was negative for regional or distant disease. Sixty-five percent of patients had a sentinel lymph node biopsy (SNLB) confirming node-negative disease. Clinical and pathologic characteristics of the patient cohort are separated by WLE margin size and summarized in Table 1. The median age was 73 years (range 32–94 years), and 65% of patients were male. Of the 79 patients, 5 (6%) had a history of immunosuppression secondary to organ transplant or autoimmune disease. The location of the tumors was as follows: 41 head and neck (52%), 16 upper extremity (20%), 12 lower extremity (15%), and 10 trunk (13%) tumors. A total of 62 tumors (79%) were ≤ 2 cm, 12 tumors (15%) were > 2 but ≤ 5 cm, and 5 tumors (6%) were > 5 cm. The median tumor depth was 4.5 mm (range 1–29 mm) and 20% of tumors demonstrated LVSI. Reduced WLE margins were not significantly associated with immune status, tumor location, tumor size, tumor depth, or the presence of LVSI (Table 1).

TABLE 1 Clinical and pathologic characteristics for each WLE margin group

WLE margin	Total (<i>n</i> = 79)	< 1 cm (<i>n</i> = 13)	1 cm ≤ <i>x</i> < 2 cm (<i>n</i> = 44)	≥ 2 cm (<i>n</i> = 22)	<i>p</i>
Age					0.17
Median (range)	73 (32–94)	78 (59–91)	73.5 (32–94)	69 (56–88)	
Gender					0.19
Male	51 (65)	10 (77)	28 (63)	13 (59)	
Female	28 (35)	3 (23)	16 (37)	9 (41)	
Immunosuppressed					0.27
Yes	5 (6)	2 (15)	1 (2)	2 (10)	
No	74 (94)	11 (85)	43 (98)	20 (90)	
Site of tumor					0.15
Head and neck	41 (52)	8 (64)	23 (52)	8 (36)	
Upper extremity	16 (20)	2 (14)	11 (25)	4 (18)	
Lower extremity	12 (15)	3 (21)	6 (14)	3 (14)	
Trunk	10 (13)	0 (0)	4 (9)	7 (32)	
Tumor size					0.10
≤ 2 cm	62 (79)	8 (62)	36 (82)	18 (82)	
2 < <i>x</i> ≤ 5 cm	12 (15)	2 (16)	7 (16)	3 (14)	
> 5 cm	5 (6)	3 (23)	1 (2)	1 (4)	
Tumor depth (cm)					0.92
Median (range)	0.45 (0.1–2.9)	0.40 (0.2–0.7)	0.47 (0.1–2.9)	0.43 (0.2–1.2)	
LVSI					0.51
Yes	16 (20)	4 (31)	8 (18)	4 (14)	
No	63 (80)	9 (69)	36 (82)	18 (86)	
Pathologic margin					0.01
Positive	5 (6)	4 (31)	0 (0)	1 (5)	
Close	12 (15)	2 (15)	9 (20)	1 (5)	
Clear	62 (78)	7 (54)	35 (80)	20 (90)	
Adjuvant radiation					0.60
Yes	25 (32)	7 (54)	10 (23)	8 (36)	
No	54 (68)	6 (46)	34 (77)	14 (64)	

LVSI lymphovascular invasion

A total of 25 patients (32%) received aRT. NCCN guidelines for 2009–2020 have recommended aRT in immunocompromised patients, tumors ≥ 1 cm, when LVSI is present, and when pathologic margins are positive.^{2–5} aRT was administered to 6 patients (26%) without any of these RT indications, 8 patients (21%) with 1 indication, 10 patients (71%) with 2 indications, and 1 patient (25%) with 3 or more indications. aRT was administered to 32% of patients with tumors ≥ 1 cm, 56% of patients with LVSI, 50% of immunocompromised patients, and 60% of patients with a positive margin.

The median follow-up time for all patients was 28.6 months (range 1.1–171.2 months). At last follow-up, 31 patients (39%) developed MCC recurrence including 11 patients with local recurrence (14%), 17 patients with regional recurrence (22%), and 10 patients (13%) with

distant recurrence. Twenty-five of the 31 patients recurred at a single site (local, regional, or distant), while 6 of 31 recurred at multiple sites. At last follow-up, 33 patients (42%) had died, including 13 patients who died from MCC (16%). For those that died from MCC, the median time from diagnosis to death was 20.8 months.

REDUCED WIDE LOCAL EXCISION MARGINS ARE ASSOCIATED WITH REDUCED DISEASE-FREE SURVIVAL AND DISEASE-SPECIFIC SURVIVAL

A total of 31 patients developed recurrent disease (39%) (Table 2). Among the 31 patients with tumor recurrence, WLEs margins of < 1 cm were achieved in 7 patients (23%), 1- to 1.9-cm margins were achieved in 19 patients

TABLE 2 Local recurrence, regional recurrence, distant recurrence, all-cause recurrence, and death from MCC for each WLE margin group

WLE margin	< 1 cm (<i>n</i> = 13)	1 cm ≤ <i>x</i> < 2 cm (<i>n</i> = 44)	≥ 2 cm (<i>n</i> = 22)	<i>p</i>
Relapse, any cause	7 (54)	19 (43)	5 (23)	0.009
Local failure	3 (23)	6 (14)	2 (9)	0.126
Regional failure	4 (31)	11 (33)	2 (9)	0.033
Distant failure	3 (23)	6 (14)	1 (5)	0.070
Death from MCC	4 (31)	8 (18)	1 (5)	0.014

(61%), and ≥ 2-cm margins were achieved in 5 patients (16%). The proportion of patients relapsing with WLE margins < 1 cm, 1–1.9 cm, and ≥ 2 cm or more was 54%, 42%, and 23%, respectively ($p = 0.009$). The proportion of patients dying from MCC with WLE margins < 1 cm, 1.0–1.9 cm, and ≥ 2 cm was 31%, 18%, and 5%, respectively ($p = 0.014$). The observation that tumor recurrence and death occurred disproportionally in patients with smaller WLE margins suggested that there might be a linear relationship between margin size and risk of relapse.

Univariable Cox regression analysis was used to evaluate how the size of WLE margins and other covariables influenced patient outcomes. We first divided patients into three subgroups based on WLE margin size: < 1 cm, $1 \leq x < 2$ cm, and ≥ 2 cm. (Figure 1). Decreased WLE margins was tightly correlated with DSS and DFS. The DFS hazard ratio increased to 1.9 and 3.0 as WLE margins decreased from ≥ 2 to 1–1.9 cm and from ≥ 2 to < 1 cm, respectively ($p = 0.04$) (Fig. 1a). Similarly, the DSS hazard ratio increased to 4.3 and 7.7 as tumor margins decreased from ≥ 2 to 1–1.9 cm and from ≥ 2 to < 1 cm, respectively ($p = 0.02$) (Fig. 1b). At 1 year, DFS was 51.3%, 71.4%, and 87.8% for patients with WLE margins < 1 cm, 1–1.9 cm, and ≥ 2 cm, respectively ($p = 0.02$). At 3 years, the DSS was 57.7%, 82.6%, and 100% for patients with WLE margins < 1 cm, 1–1.9 cm,

and ≥ 2 cm, respectively ($p = 0.02$). Together, this suggested that the risk of recurrence and death from MCC are both sensitive to the size of WLE margins. To further quantify the risk of recurrence or death from MCC for a finite change in WLE margins, we next evaluated WLE margins as a continuous variable (Table 3). Every 1-cm increase in WLE margin was associated with improved LRFS (HR 0.35, 95% CI 0.11–1.13, $p = 0.07$), RRFS (HR 0.33, 95% CI 0.12–0.86, $p = 0.02$), DRFS (HR 0.30, 95% CI 0.08–0.99, $p = 0.049$), DFS (HR 0.42, 95% CI 0.22–0.82, $p = 0.01$), and DSS (HR 0.19, 95% CI 0.05–0.65, $p = 0.008$). Several additional covariates were associated with poorer prognosis including immunosuppression, LVSI, and male gender (Table 3). However, adjuvant radiation was not significantly associated with DFS or DSS.

Multivariable analysis demonstrated that increasing WLE margins was associated with improved RRFS (HR 0.28, 95% CI 0.11–0.75, $p = 0.01$), DRFS (HR 0.30, 95% CI 0.08–0.99, $p = 0.049$), DFS (HR 0.42, 95% CI 0.21–0.86, $p = 0.02$), and DSS (HR 0.16, 95% CI 0.04–0.61, $p = 0.007$) (Table 3). Thus, every 1-cm increase in the size of WLE margins was associated with a 72% lower risk of regional failure, 70% lower risk of distant failure, 58% lower risk of disease recurrence, and 84% lower risk of death from MCC. Several additional

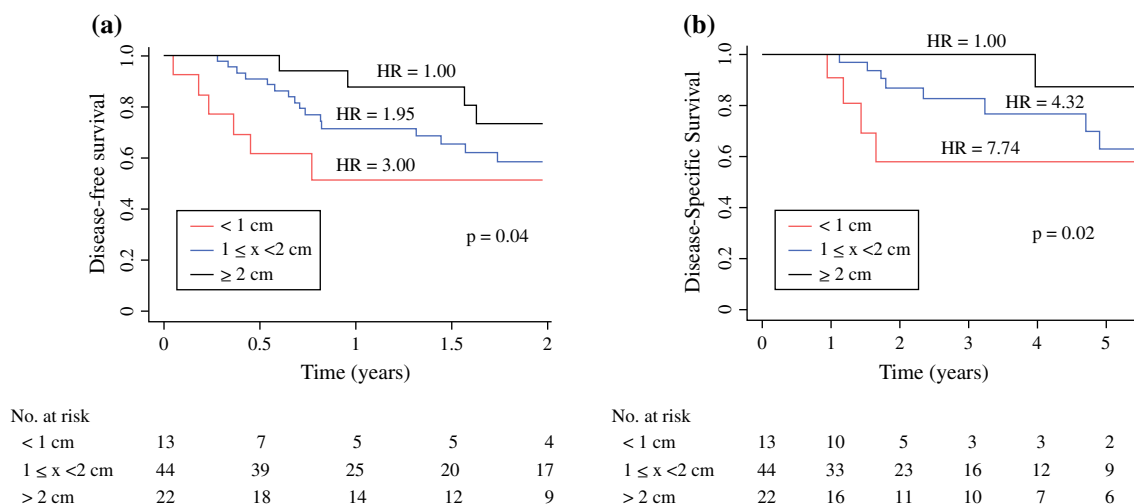
**FIG. 1** Kaplan-Meier plots showing the effect of WLE margin size on (a) disease-free survival (DFS) and (b) disease-specific survival (DSS)

TABLE 3 Univariable and multivariable cox regression analysis of WLE margins and pathologic margins

	LRFS		RRFS		DRFS		DFS		DSS	
	HR [CI]	<i>p</i>	HR [CI]	<i>p</i>	HR [CI]	<i>p</i>	HR [CI]	<i>p</i>	HR [CI]	<i>p</i>
Univariable analysis										
WLE margins	0.35 [0.11–1.13]	0.07	0.33* [0.12–0.86]	0.02	0.30* [0.08–0.99]	0.049	0.42* [0.22–0.82]	0.01	0.19* [0.05–0.65]	0.008
Path margins (close/positive)	2.52 [0.73–8.73]	0.14	2.91* [1.13–7.49]	0.004	6.83* [1.80–25.9]	0.005	3.98* [1.65–4.03]	1×10^{-4}	5.06* [1.68–15.3]	0.004
Age (> 75 y/o)	1.58 [0.48–5.21]	0.44	1.55 [0.47–5.09]	0.47	1.56 [0.43–5.61]	0.50	1.57 [0.78–3.15]	0.21	2.52 [0.9–17.2]	0.23
Gender (M vs F)	5.59 [0.71–43.7]	0.10	5.09* [1.17–22.1]	0.03	2.26 [0.48–10.7]	0.30	3.46* [1.33–8.96]	0.01	6.59 [0.86–50.7]	0.07
Immune compromised	6.75* [1.77–25.7]	.005	1.04 [0.14–7.87]	0.97	2.35 [0.29–19.1]	0.42	4.25* [1.62–11.1]	0.003	1.31 [0.81–2.01]	0.20
Disease site		0.54		0.20		0.85		0.26		0.56
Trunk	Ref		Ref		Ref		Ref		Ref	
H&N	1.53 [0.18–12.7]		0.65 [0.13–1.55]		1.30 [0.15–11.2]		0.60 [0.23–1.57]		0.85 [0.17–4.26]	
Extremity	1.45 [0.16–13.0]		0.61 [0.18–2.07]		1.29 [0.14–11.6]		0.75 [0.28–1.99]		0.83 [0.16–4.29]	
LVSI	5.15* [1.55–17.1]	0.007	3.46* [1.30–9.20]	0.01	0.91 [0.11–7.29]	0.93	3.55* [1.62–7.80]	0.002	3.37* [1.02–11.1]	0.04
aRT	1.54 [0.43–5.54]	0.12	1.84 [0.56–6.04]	0.31	1.62 [0.43–6.15]	0.48	1.47 [0.71–3.04]	0.30	2.11 [0.64–6.98]	0.22
Multivariable analysis of WLE margins										
WLE margins	0.44 [0.13–1.48]	0.18	0.28* [0.11–0.75]	0.01	0.30* [0.08–0.99]	0.049	0.42* [0.21–0.86]	0.02	0.16* [0.04–0.61]	0.007
Gender (M vs F)	8.13 [0.96–68.6]	0.05	5.16* [1.14–23.3]	0.03	–		5.86* [1.9–17.8]	0.002	6.99 [0.89–54.3]	0.06
Immune compromised	4.59* [1.08–19.5]	0.04	–		–		1.78 [0.55–5.7]	0.33	–	
LVSI	5.04* [1.43–17.7]	0.01	6.21* [2.05–18.8]	0.001	–		4.18* [1.6–9.7]	0.003	3.66* [1.05–12.7]	0.04
Wald statistic		0.005		9×10^{-4}				3×10^{-5}		0.02
Multivariable analysis of pathologic margins										
Path margins (close/positive)	–	–	2.08 [0.75–5.32]	0.16	6.83* [1.80–25.9]	0.005	2.98* [1.31–6.75]	0.009	3.52* [1.14–10.9]	0.02
Gender (M vs F)	6.05 [0.77–47.4]	0.08	4.88* [1.11–21.4]	0.04	–		3.04* [1.13–8.11]	0.03	5.20 [0.66–41.1]	0.11
Immune compromised	3.79 [0.90–16.0]	0.07	–		–		1.12 [0.41–5.43]	0.84	–	
LVSI	3.69* [1.02–13.3]	0.04	3.31* [1.19–9.14]	0.02	–		3.25* [1.37–7.71]	0.007	2.94 [0.85–10.1]	0.08
Wald statistic		0.004		0.003		0.005		4×10^{-5}		0.005

HR hazard ratio; CI 95% confidence interval; LRFS local recurrence-free survival; RRFS regional recurrence-free survival; DRFS distant recurrence-free survival; DFS disease-free survival; DSS disease-specific survival

* $p < 0.05$

covariates remained significant on multivariable analysis. Immunosuppression was associated with reduced LRFS (HR 4.59, 95% CI 1.08–19.5, $p = 0.04$). Male gender was associated with reduced DFS (HR 5.86, 95% CI 1.90–17.8, $p = 0.002$) and DSS (HR 6.99, 95% CI 0.89–54.3, $p = 0.06$), which trended toward significance. LVSI was significantly associated with poorer prognosis including reduced LRFS (HR 5.04, 95% CI 1.43–17.7, $p = 0.01$), RRFS (HR 6.21, 95% CI 2.05–18.8, $p = 0.001$), DFS (HR 4.18, 95% CI 1.6–9.7, $p = 0.003$), and DSS (HR 3.66, 95% CI 1.05–12.7, $p = 0.04$).

Close or Positive Pathological Margins are Associated with Increased Risk of Relapse and Death from MCC

We next investigated whether reduced WLE margins were associated with an increased risk of close/positive margins, which was defined as peripheral or deep margins < 3 mm. WLE margins and pathologic margins were moderately-to-strongly correlated (Pearson correlation coefficient = 0.66). As shown in Table 1, there was a significant increase in pathologic close/positive margins in patients with reduced WLE margins ($p = 0.01$). In contrast, there was not a significant increase in close/positive margins in immunosuppressed patients ($p = 0.24$) or patients with LVSI ($p = 0.16$), higher clinical T stage ($p = 0.28$), or greater depth of invasion ($p = 0.34$). Given that reduced WLE margins at the time of surgery were associated with poorer oncologic outcomes and correlated with reduced size of pathologic margins, we next assessed whether reduced pathologic margins were associated with poorer outcomes. Close/positive margins were associated with reduced LRFS (HR 2.52, 95% CI 0.73–8.73, $p = 0.14$), but this failed to reach significance. Close/positive margins were associated with significantly reduced RRFS (HR 2.91, 95% CI 1.13–7.49, $p = 0.004$), DRFS (HR 6.83, 95% CI 1.80–25.9, $p = 0.005$), DFS (HR 3.98, 95% CI 1.65–4.03, $p < 0.001$), and DSS (HR 5.06, 95% CI 1.68–15.3, $p = 0.004$). Multivariable analysis demonstrated that close/positive pathologic margins were associated with reduced DRFS (HR 6.83, 95% CI 1.80–25.9, $p = 0.005$), DFS (HR 2.98, 95% CI 1.31–6.75, $p = 0.009$) (Fig. 2a), and DSS (HR 3.52, 95% CI 1.14–10.9, $p = 0.02$) (Fig. 2b). These results are summarized in Table 3.

DISCUSSION

Here we show that reduced WLE margins are associated with smaller pathologic margins, and both are associated with increased risk of relapse and death from Merkel cell carcinoma. For every 1-cm increase in the size of WLE margins, there was a 58% lower risk of disease-recurrence

and 84% lower risk of death from MCC. Consistent with prior studies, LVSI, immunosuppression, and male gender were associated with poorer outcomes.^{11–14} When WLE margins are close or positive, less aggressive tumors have a higher propensity to recur locally.^{15–17} In contrast, Merkel cell carcinoma is highly aggressive, with very small tumors demonstrating a high propensity to metastasize to regional lymphatics and distant sites including soft tissue, visceral organs, and bones. Consistent with this, tumors in our cohort were small, with a median tumor size of 1.4 cm and 85% of tumors less than 3 cm, but with a reported regional and distant relapse rate of 22% and 13%, respectively. Our study addresses an important question that has not been previously investigated, by extending our analysis to regional and distant relapse.

Due to the rarity of MCC, optimal WLE margins are not well defined. Our study results differ from the two other retrospective studies from Memorial Sloan-Kettering and Moffitt.^{6,7} The largest study, by Perez et al., showed that there was no significant difference in local recurrence rates between 1- and 2-cm WLE margins.⁷ In contrast, our study showed that recurrence risk increases with reduced WLE margins. Given the rarity of MCC, studies frequently consist of small sample sizes, and are susceptible to Type II errors, which may explain why our study showed a difference, while others did not. In addition, the proportion of patients who received aRT was very different between these studies. In Perez et al., approximately 70% of patients received adjuvant radiation, compared to only 32% in our study, and radiation treatment may have diluted out any significant differences in recurrence rates between different margin groups. Our study, as well as limited data in the literature,^{10,18} strongly supports that WLE size matters in the absence of aRT.

The need for larger WLE margins may be less important when aRT is administered.¹⁸ MCC is a highly radiosensitive tumor, and current NCCN guidelines recommend aRT in patients with tumors ≥ 1 cm or in the presence of additional risk factors (immunosuppression, LVSI, positive margins).² While large retrospective studies have shown that aRT improves overall survival in localized MCC,^{9,19–22} our study failed to show that aRT was associated with improved DFS and DSS. This could be due to several reasons. First, the overall number of patients who received aRT was comparatively lower when compared with other studies.^{7,10,18} Seventy-four percent of patients in our study had at least one indication for aRT (tumor ≥ 1 cm, LVSI, positive margin), but only 32% of patients received aRT. Second, the retrospective nature of this study introduces an inherent selection bias, where many of those selected to receive aRT were patients with more aggressive disease associated with a worse prognosis. While many advocate for reduced surgical margins in the

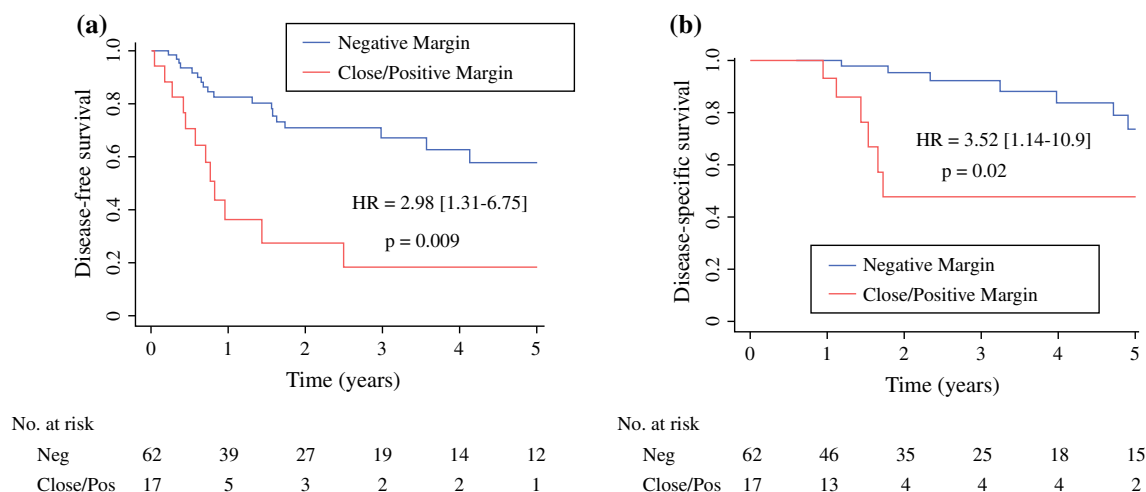


FIG. 2 Kaplan-Meier plots showing the effect of close/positive margins versus negative margins on (a) disease-free survival (DFS), and (b) disease-specific survival (DSS)

setting of adjuvant radiation, there are several practical considerations. First, at the time of surgery, key pieces of information are missing (i.e., LVSI and SLNB status), which are needed to determine whether aRT is indicated. Second, many patients with indications for aRT fail to receive aRT. Third, delays in initiating radiotherapy after surgery are associated with worse outcomes in MCC and other cancers,^{23,24} and tend to occur in patients who exhibit poor postoperative wound healing or experience delays in referral to radiation oncology. Thus, in practice, radiation may not fully compensate for reduced surgical margins, as is evident from our patient cohort; however, the reasons for this may not be directly related to the value of radiotherapy itself but in the initiation of treatment relative to surgery, as has been shown for other cancers.^{25,26}

This study has several weaknesses. First, it is a retrospective study from a single institution. Unfortunately, prospective trials to address the question of surgical margins are lacking, and current guidelines are based solely on similar retrospective data series. As a referral center, the possibility for a selection bias exists, as often patients with more advanced disease are referred, and this may explain the higher incidence of recurrence in our patient cohort (38%). Second, reduced WLE and pathologic margins were not associated with adverse clinicopathologic features (tumor size, LVSI, immune suppression), suggesting that smaller margins are not a surrogate for more aggressive disease, but the small sample size of this study limits our ability to fully address this question. Third, while all patients had preoperative imaging demonstrating no evidence of nodal involvement, only 65% of patients had a sentinel lymph node biopsy to confirm node-negative disease. Given that the false-negative rates of preoperative imaging can approach 30%, it is plausible that clinically

node-negative patients could have had pathological nodal involvement at diagnosis that was missed on preoperative imaging. Ten of the 30 patients that developed relapse failed to have a SNLB. Three of the 10 patients developed an isolated local failure. Thus, it is plausible that 2–3 of the 7 remaining patients could have had nodal involvement at diagnosis which was undetected on imaging. Importantly, these 7 patients were equally distributed amongst the surgical margin groups (2, 2, 3), so it should not influence the results of this study.

In summary, this risk of recurrence and death increases with reduced WLE margins, suggesting that surgical resection should aim for 2-cm margins when feasible, especially if it does not compromise the functional and cosmetic priorities of the patient, or in situations where aRT is not likely. Formally establishing the optimum excisional margin for MCC patients requires a multicenter randomized trial where outcomes could be assessed by prospectively assigning patients to either 1-cm or 2-cm excisional margins, with postoperative adjuvant radiation applied across both arms using predefined criteria. While high level clinical evidence is lacking, we believe our study should raise concern that reducing margins may lead to poorer outcomes, especially in the absence of postoperative radiotherapy.

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DISCLOSURE The authors do not have any significant conflicts of interest to report.

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